

Section 13.3

Series Solution of Special Function

13.3.1 Some Special Functions

(i) **Bessel's differential equations:**

The differential equation of the form

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0$$

is called Bessel's differential equation of order n , where n is a non-negative constant.

(ii) **Legendre's differential equation:**

The differential equation of the form

$$(1-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$$

is called Legendre's differential equation of order n , where n is a positive integer.

(iii) **Hermite's differential equation:**

A differential equation of the form

$$\frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2ny = 0$$

is called Hermite's differential equation of order n , where n is a constant.

(iv) **Laguerre's differential equation:**

A differential equation of the form

$$x \frac{d^2 y}{dx^2} + (1-x) \frac{dy}{dx} + ny = 0$$

is called Laguerre's differential equation of order n , where n is a non-negative integer.

Solved Problems

Example-1 Solve Bessel's differential equation $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0$ by the method of Frobenius,

when n is a non-negative integer constant.

Solution Given differential equation is

$$x^2 y'' + xy' + (x^2 - n^2)y = 0 \quad \dots(1)$$

Let the trial solution of (1) be

$$y = \sum_{r=0}^{\infty} a_r x^{c+r}, \quad a_0 \neq 0 \quad \dots (2)$$

$$\therefore y' = \sum_{r=0}^{\infty} a_r (c+r) x^{c+r-1}$$

$$\text{and } y'' = \sum_{r=0}^{\infty} a_r (c+r) (c+r-1) x^{c+r-2}$$

Now putting the values of y, y', y'' in (1) we get

$$\begin{aligned} & x^2 \sum_{r=0}^{\infty} a_r (c+r) (c+r-1) x^{c+r-2} + x \sum_{r=0}^{\infty} a_r (c+r) x^{c+r-1} + (x^2 - n^2) \sum_{r=0}^{\infty} a_r x^{c+r} = 0 \\ \Rightarrow & \sum_{r=0}^{\infty} a_r (c+r) (c+r-1) x^{c+r} + \sum_{r=0}^{\infty} a_r (c+r) x^{c+r} + \sum_{r=0}^{\infty} a_r x^{c+r+2} - \sum_{r=0}^{\infty} n^2 a_r x^{c+r} = 0 \\ \Rightarrow & \sum_{r=0}^{\infty} a_r [(c+r)^2 - (c+r) + (c+r) - n^2] x^{c+r} + \sum_{r=0}^{\infty} a_r x^{c+r+2} = 0 \\ \Rightarrow & \sum_{r=0}^{\infty} a_r [(c+r)^2 - n^2] x^{c+r} + \sum_{r=2}^{\infty} a_{r-2} x^{c+r} = 0 \\ \Rightarrow & a_0 (c^2 - n^2) x^c + a_1 [(c+1)^2 - n^2] x^{c+1} + \sum_{r=2}^{\infty} a_r [(c+r)^2 - n^2] x^{c+r} + \sum_{r=2}^{\infty} a_{r-2} x^{c+r} = 0 \\ \Rightarrow & a_0 (c^2 - n^2) x^c + a_1 [(c+1)^2 - n^2] x^{c+1} + \sum_{r=2}^{\infty} [a_r \{(c+r)^2 - n^2\} + a_{r-2}] x^{c+r} = 0 \end{aligned}$$

Equating to zero the coefficient of lowest power of x (i.e. coefficient of x^c) then the indicial equation is

$$\begin{aligned} & a_0 (c^2 - n^2) = 0 \\ \Rightarrow & c^2 - n^2 = 0 \quad [\text{since } a_0 \neq 0] \\ \therefore & c = n, -n \end{aligned}$$

Also equating to zero the coefficient of x^{c+1} , we get

$$\begin{aligned} & a_1 [(c+1)^2 - n^2] = 0 \\ \Rightarrow & a_1 = 0 \quad [\text{since } (c+1)^2 - n^2 \neq 0, \text{ for } c = n, -n] \end{aligned}$$

Again equating to zero the coefficient of x^{c+r} , then the recurrence relation is

$$\begin{aligned} & a_r [(c+r)^2 - n^2] + a_{r-2} = 0 \\ \Rightarrow & a_r = \frac{-a_{r-2}}{(c+r)^2 - n^2} \\ \Rightarrow & a_r = \frac{-a_{r-2}}{(c+r-n)(c+r+n)}; r \geq 2 \quad \dots (3) \end{aligned}$$

Now putting $r=2, 3, 4, 5, 6, \dots$ in (3), we get

$$a_2 = \frac{-a_0}{(c+2-n)(c+2+n)}$$

$$a_3 = \frac{-a_1}{(c+3-n)(c+3+n)} = 0 \quad [\text{since } a_1 = 0]$$

$$a_4 = \frac{-a_2}{(c+4-n)(c+4+n)} = \frac{a_0}{(c+4-n)(c+4+n)(c+2-n)(c+2+n)}$$

$$a_5 = \frac{-a_3}{(c+5-n)(c+5+n)} = 0 \quad [\text{since } a_3 = 0]$$

and so on.

$$\text{and } a_1 = a_3 = a_5 = a_7 = \dots = 0$$

From (2) we get

$$y = x^c [a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \dots]$$

$$\Rightarrow y = x^c \left[a_0 + 0 - \frac{a_0 x^2}{(c+2-n)(c+2+n)} + 0 + \frac{a_0 x^4}{(c+4-n)(c+4+n)(c+2-n)(c+2+n)} - \dots \right]$$

$$\Rightarrow y = a_0 x^c \left[1 - \frac{x^2}{(c+2-n)(c+2+n)} + \frac{x^4}{(c+2-n)(c+4-n)(c+2+n)(c+4+n)} - \dots \right]$$

$$\text{Let } y(x, c) = a_0 x^c \left[1 - \frac{x^2}{(c+2-n)(c+2+n)} + \frac{x^4}{(c+2-n)(c+4-n)(c+2+n)(c+4+n)} - \dots \right] \dots (4)$$

Now putting $c = n$ and $c = -n$ in (4) successively then the two independent solutions y_1 and y_2 are obtained

$$y_1 = [y(x, c)]_{c=n} = a_0 x^n \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4(2n+2)(2n+4)} - \dots \right] \dots (5)$$

$$\Rightarrow y_1 = a_0 x^n \left[1 - \frac{x^2}{2^2(n+1)} + \frac{x^4}{2^2 \cdot 2^3(n+1)(n+2)} - \dots \right], \text{ provided that } n \text{ is non negative integer.}$$

$$\text{and } y_2 = [y(x, c)]_{c=-n} = a_0 x^{-n} \left[1 - \frac{x^2}{(2-2n)2} + \frac{x^4}{(2-2n)(4-2n)2 \cdot 4} - \dots \right] \dots (6)$$

$$\Rightarrow y_2 = a_0 x^{-n} \left[1 - \frac{x^2}{2^2(1-n)} + \frac{x^4}{2^2 \cdot 2^3(1-n)(2-n)} - \dots \right], \text{ provided that } n \text{ is non negative integer.}$$

If we choose $a_0 = 1$, then the required solution is

$$y = c_1 y_1 + c_2 y_2$$

where c_1 and c_2 are arbitrary constants. (**Ans.**)

Alternative form of the solution:

In defining Bessel functions it is usual to write a_0 as

$$a_0 = \frac{1}{2^n \Gamma(n+1)}$$

Substituting this value of a_0 in (5) define the Bessel function of the first kind and order n , $J_n(x)$, such that

$$y = J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \quad \dots (7)$$

Similarly we find second solution from (6) as

$$y = J_{-n}(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(-n+r+1)} \left(\frac{x}{2}\right)^{-n+2r} \quad \dots (8)$$

The general solution of Bessel's equation for non- integral n is therefore

$$y = AJ_n(x) + BJ_{-n}(x) \quad \dots (9)$$

where A and B are arbitrary constants.

When n is an positive integer:

In this case we can write $\Gamma(n+r+1) = (n+r)!$. Then we can write (7) and (8) as

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r!(n+r)!} \left(\frac{x}{2}\right)^{n+2r} \quad \dots (10)$$

$$J_{-n}(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r!(-n+r)!} \left(\frac{x}{2}\right)^{-n+2r} \quad \dots (11)$$

Now the first n terms of the series (11) are zero since Γ - function or factorial function for negative integers is infinite. Hence putting $r = n + p$ in (11), we have

$$\begin{aligned} J_{-n}(x) &= \sum_{r=n}^{\infty} \frac{(-1)^r}{r!(-n+r)!} \left(\frac{x}{2}\right)^{-n+2r} \\ &= \sum_{p=0}^{\infty} \frac{(-1)^{n+p}}{p!(n+p)!} \left(\frac{x}{2}\right)^{n+2p} \\ &= (-1)^n J_n(x) \end{aligned}$$

We see then that $J_n(x)$ and $J_{-n}(x)$ are linearly dependent. As a result the Frobenius method gives only one solution in this circumstance, and therefore (9) cannot be taken as the general solution of the Bessel equation when n is an positive integer. A second solution, however, may be obtained by using another method.

Example-2 Solve the following Bessel's differential equation of order zero:

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + xy = 0 .$$

Solution Bessel's differential equation of order zero is given by

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + xy = 0 \quad \dots(1)$$

We observe that $x=0$ is a regular singular point. We shall seek a solution of equation (1) which are valid in an interval $0 < x < R$. Let the trial solution of (1) be

$$y = \sum_{r=0}^{\infty} a_r x^{c+r}, \quad a_0 \neq 0 \quad \dots (2)$$

$$\therefore \frac{dy}{dx} = \sum_{r=0}^{\infty} a_r (c+r) x^{c+r-1}$$

$$\text{and } \frac{d^2 y}{dx^2} = \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2}$$

Now putting the values of $y, \frac{dy}{dx}, \frac{d^2 y}{dx^2}$ in (1), we get

$$\begin{aligned} & x \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} + \sum_{r=0}^{\infty} a_r (c+r) x^{c+r-1} + x \sum_{r=0}^{\infty} a_r x^{c+r} = 0 \\ \Rightarrow & \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-1} + \sum_{r=0}^{\infty} a_r (c+r) x^{c+r-1} + \sum_{r=0}^{\infty} a_r x^{c+r+1} = 0 \\ \Rightarrow & \sum_{r=0}^{\infty} a_r (c+r) \{(c+r)-1+1\} x^{c+r-1} + \sum_{r=2}^{\infty} a_{r-2} x^{c+r-1} = 0 \\ \Rightarrow & \sum_{r=0}^{\infty} a_r (c+r)^2 x^{c+r-1} + \sum_{r=2}^{\infty} a_{r-2} x^{c+r-1} = 0 \\ \Rightarrow & a_0 c^2 x^{c-1} + a_1 (c+1)^2 x^c + \sum_{r=2}^{\infty} a_r (c+r)^2 x^{c+r-1} + \sum_{r=2}^{\infty} a_{r-2} x^{c+r-1} = 0 \\ \Rightarrow & a_0 c^2 x^{c-1} + a_1 (c+1)^2 x^c + \sum_{r=2}^{\infty} \{a_r (c+r)^2 + a_{r-2}\} x^{c+r-1} = 0 \end{aligned}$$

Equating to zero the coefficient of lowest power of x (i.e. coefficient of x^{c-1}) then the indicial equation is

$$a_0 c^2 = 0$$

$$\Rightarrow c^2 = 0 \quad [\text{since } a_0 \neq 0]$$

$$\therefore c = 0, 0$$

Also equating to zero the coefficient of x^c , we get

$$a_1 (c+1)^2 = 0$$

$$\Rightarrow a_1 = 0 \quad [\text{since } (c+1)^2 \neq 0, \text{ for } c = 0]$$

Again equating to zero the coefficient of x^{c+r-1} , then the recurrence relation is

$$a_r(c+r)^2 + a_{r-2} = 0$$

$$\Rightarrow a_r = \frac{-a_{r-2}}{(c+r)^2}; r \geq 2 \quad \dots (3)$$

For $c = 0$, the recurrence relation becomes

$$\Rightarrow a_r = \frac{-a_{r-2}}{r^2}; r \geq 2 \quad \dots (4)$$

Now putting $r = 2, 3, 4, 5, 6, \dots$ in (4), we get

$$a_2 = \frac{-a_0}{2^2}$$

$$a_3 = \frac{-a_1}{3^2} = 0 \quad [\text{since } a_1 = 0]$$

$$a_4 = \frac{-a_2}{4^2} = \frac{-1}{4^2} \cdot \frac{-a_0}{2^2} = \frac{a_0}{2^2 \cdot 4^2}$$

$$a_5 = \frac{-a_3}{5^2} = 0 \quad [\text{since } a_3 = 0]$$

and so on.

$$\text{and } a_1 = a_3 = a_5 = a_7 = \dots = 0$$

The general even coefficient may be written as

$$a_{2r} = \frac{(-1)^r a_0}{2^2 \cdot 4^2 \cdot 6^2 \cdot \dots \cdot (2r)^2} = \frac{(-1)^r a_0}{(r!)^2 \cdot 2^{2r}}; \quad r \geq 1$$

Thus for $c = 0$, one solution of equation (1) is obtain from equation (2) using the values of a_{2r}

$$y_1(x) = a_0 \sum_{r=0}^{\infty} \frac{(-1)^r}{(r!)^2} \left(\frac{x}{2}\right)^{2r}$$

If we set $a_0 = 1$, then we obtain a particular solution of (1) which is denoted by $J_0(x)$, and called the **Bessel function of first kind of order zero**, is defined by

$$J_0(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{(r!)^2} \left(\frac{x}{2}\right)^{2r} \quad \dots (5)$$

Since the roots of the indicial equation are equal, we know that a solution of (1), which is linearly independent of J_0 , must be of the form

$$y = x \sum_{r=0}^{\infty} a_r^* x^r + J_0(x) \ln x$$

for the interval $0 < x < R$.

We know if f is a nontrivial solution of the second order differential equation

$$a_0(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_2(x)y = 0 \quad \dots (6)$$

then $g(x) = f(x)v(x)$ is a solution of the equation (6) by the method of reduction of order, where $v(x)$ is defined by

$$v(x) = \int \frac{e^{-\int \frac{a_1(x)}{a_0(x)} dx}}{[f(x)]^2} dx$$

Let the independent solution is given by

$$\begin{aligned} y_2(x) &= J_0(x) \int \frac{e^{-\int \frac{dx}{x}}}{[J_0(x)]^2} dx \\ \Rightarrow y_2(x) &= J_0(x) \int \frac{dx}{x[J_0(x)]^2} \quad \dots (7) \end{aligned}$$

Few terms of the series (5) is written as

$$\begin{aligned} J_0(x) &= 1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + \dots \\ \Rightarrow [J_0(x)]^2 &= \left(1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + \dots\right)^2 = 1 + \left(-\frac{x^2}{4}\right)^2 \\ &\quad + 2 \cdot 1 \cdot \left(-\frac{x^2}{4}\right) + 2 \cdot 1 \cdot \frac{x^4}{64} + 2 \cdot 1 \cdot \left(-\frac{x^6}{2304}\right) + 2 \cdot \left(-\frac{x^2}{4}\right) \cdot \frac{x^4}{64} + \dots \\ \Rightarrow [J_0(x)]^2 &= 1 - \frac{x^2}{2} + \frac{3x^4}{32} - \frac{5x^6}{576} + \dots \\ \Rightarrow \frac{1}{[J_0(x)]^2} &= \left\{1 - \left(\frac{x^2}{2} - \frac{3x^4}{32} + \frac{5x^6}{576} - \dots\right)\right\}^{-1} \\ \Rightarrow \frac{1}{[J_0(x)]^2} &= 1 + \frac{x^2}{2} + \frac{5x^4}{32} + \frac{23x^6}{576} + \dots \end{aligned}$$

Thus from (7), we obtain

$$\begin{aligned} y_2(x) &= J_0(x) \int \left(\frac{1}{x} + \frac{x}{2} + \frac{5x^3}{32} + \frac{23x^5}{576} + \dots \right) dx \\ &= J_0(x) \left(\ln x + \frac{x^2}{4} + \frac{5x^4}{128} + \frac{23x^6}{3456} + \dots \right) \\ &= J_0(x) \ln x + \left(1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + \dots \right) \left(\frac{x^2}{4} + \frac{5x^4}{128} + \frac{23x^6}{3456} + \dots \right) \\ &= J_0(x) \ln x + \frac{x^2}{4} - \frac{3x^4}{128} + \frac{11x^6}{13824} + \dots \end{aligned}$$

$$= J_0(x) \ln x + \frac{x^2}{2^2} - \frac{x^4}{2^4(2!)^2} \left(1 + \frac{1}{2}\right) + \frac{x^6}{2^6(3!)^2} \left(1 + \frac{1}{2} + \frac{1}{3}\right) + \dots$$

Since $(-1)^2 \frac{1}{2^2(1!)^2} (1) = \frac{1}{2^2} = \frac{1}{4}$

$$(-1)^3 \frac{1}{2^4(2!)^2} \left(1 + \frac{1}{2}\right) = -\frac{3}{2^4 \cdot 2^2 \cdot 2} = -\frac{3}{128}$$

$$(-1)^4 \frac{1}{2^6(3!)^2} \left(1 + \frac{1}{2} + \frac{1}{3}\right) = \frac{11}{2^6 \cdot 6^2 \cdot 6} = \frac{11}{13824}$$

Further we would certainly write the general coefficient a_{2r}^* as

$$a_{2r}^* = \frac{(-1)^{r+1}}{2^{2r}(r!)^2} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{r}\right), \quad r \geq 1$$

So $y_2(x)$ can write in the following form

$$y_2(x) = J_0(x) \ln x + \sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{2^{2r}(r!)^2} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{r}\right) \quad \dots (8)$$

Since the solution $y_2(x)$ defined by (8) is linearly independent of J_0 , we could write the general solution of equation (1) as a general linear combination of y_2 and J_0 . We choose a special linear combination denoted by Y_0 and is called **Bessel function of the second kind of order zero** and defined by

$$Y_0(x) = \frac{2}{\pi} [y_2(x) + (\gamma - \ln 2)J_0(x)] \quad \dots (9)$$

where γ is a number called *Euler's constant* and is defined by

$$\gamma = \lim_{r \rightarrow \infty} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{r} - \ln r\right) \approx 0.5772.$$

Thus the second solution is write as the function $Y_0(x)$, where

$$Y_0(x) = \frac{2}{\pi} \left[J_0(x) \ln x + \sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{2^{2r}(r!)^2} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{r}\right) + (\gamma - \ln 2)J_0(x) \right]$$

or $Y_0(x) = \frac{2}{\pi} \left[\left(\ln \frac{x}{2} + \gamma \right) J_0(x) + \sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{2^{2r}(r!)^2} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{r}\right) \right] \quad \dots (10)$

Therefore, the general solution of Bessel's equation of order zero is

$$y = c_1 J_0(x) + c_2 Y_0(x),$$

where c_1 and c_2 are arbitrary constants. (**Ans.**)

Example-3 Find two solutions of Bessel's equation of order $\frac{1}{2}$ i.e., the equation

$$x^2 y'' + xy' + \left(x^2 - \frac{1}{4}\right)y = 0; \quad 0 < x < \infty.$$

Solution Given differential equation is

$$x^2 y'' + xy' + \left(x^2 - \frac{1}{4}\right)y = 0 \quad \dots(1)$$

may be written as

$$y'' + \frac{1}{x}y' + \frac{1}{x^2}\left(x^2 - \frac{1}{4}\right)y = 0$$

This equation has a regular singular point at $x = 0$.

Since $xP(x) = 1$ and $x^2Q(x) = x^2 - \frac{1}{4}$ are both (finite) analytic at $x = 0$

$$\text{Let } y(x) = \sum_{n=0}^{\infty} a_n x^{c+n}, \quad a \neq 0 \quad \dots (2)$$

$$\therefore y'(x) = \sum_{n=0}^{\infty} (c+n)a_n x^{c+n-1} \quad \dots (3)$$

$$\text{and } y''(x) = \sum_{n=0}^{\infty} (c+n)(c+n-1)a_n x^{c+n-2} \quad \dots (4)$$

Substituting values of y , y' and y'' in (1), we get

$$\sum_{n=0}^{\infty} (n+c)(n+c-1)a_n x^{n+c} + \sum_{n=0}^{\infty} (n+c)a_n x^{n+c} - \frac{1}{4} \sum_{n=0}^{\infty} a_n x^{n+c} + \sum_{n=0}^{\infty} a_n x^{n+c+2} = 0$$

$$\text{or } \sum_{n=0}^{\infty} \left[(n+c)(n+c-1) + (n+c) - \frac{1}{4} \right] a_n x^{n+c} + \sum_{n=2}^{\infty} a_{n-2} x^{n+c} = 0$$

$$\text{or } \sum_{n=0}^{\infty} \left[(n+c)^2 - \frac{1}{4} \right] a_n x^{n+c} + \sum_{n=2}^{\infty} a_{n-2} x^{n+c} = 0$$

$$\text{or } \left(c^2 - \frac{1}{4}\right)a_0 x^c + \left[(1+c)^2 - \frac{1}{4}\right]a_1 x^{c+1} + \sum_{n=2}^{\infty} \left[\left\{ (n+c)^2 - \frac{1}{4} \right\} a_n + a_{n-2} \right] x^{n+c} = 0$$

Setting the sum of coefficients of like powers equal to zero gives

$$(i) \quad \left(c^2 - \frac{1}{4}\right)a_0 = 0$$

$$(ii) \quad \left[(1+c)^2 - \frac{1}{4}\right]a_1 = 0$$

$$\text{and } (iii) \quad \left[(n+c)^2 - \frac{1}{4}\right]a_n = -a_{n-2}; \quad n \geq 2$$

Equation (i) is the indicial equation which gives

$$c_1 = \frac{1}{2}, c_2 = -\frac{1}{2} \quad [\because a_0 \neq 0]$$

Use of $c_1 = \frac{1}{2}$ in (ii) gives $a_1 = 0$ and (iii) becomes

$$a_n = -\frac{a_{n-2}}{n(n+1)}, \quad n \geq 2 \quad \dots (5)$$

Putting $n = 2, 3, 4, \dots$ in (5) yields

$$a_2 = -\frac{a_0}{2 \cdot 3} = -\frac{1}{3!}a_0, \text{ here } a_0 \text{ is arbitrary.}$$

$$a_4 = -\frac{a_2}{4 \cdot 5} = \frac{1}{5!}a_0$$

$$a_6 = -\frac{a_4}{6 \cdot 7} = -\frac{1}{7!}a_0$$

... ..

$$a_{2n} = \frac{(-1)^n}{(2n+1)!}a_0$$

and $a_1 = a_3 = a_5 = \dots = 0$

Hence $y_1(x) = x^{1/2} \left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \dots \right)$, when $a_0 = 1$

$$= \frac{1}{\sqrt{x}} \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right) = \frac{1}{\sqrt{x}} \sin x \text{ is first solution.}$$

When $c_2 = -\frac{1}{2}$, (ii) gives $a_1 = 0$ and (iii) then becomes

$$a_n = -\frac{a_{n-2}}{n(n-1)}; \quad n \geq 2 \quad \dots (6)$$

Putting $n = 2, 3, 4, \dots$ in (6) yields

$$a_2 = \frac{-a_0}{2 \cdot 1} = -\frac{1}{2!}a_0, \quad a_0 \text{ is arbitrary.}$$

$$a_4 = \frac{-a_2}{3 \cdot 4} = \frac{1}{4!}a_0$$

$$a_6 = \frac{-a_4}{5 \cdot 6} = -\frac{1}{6!}a_0$$

... ..

$$a_{2n} = \frac{(-1)^n}{(2n)!}a_0$$

and $a_1 = a_3 = a_5 = \dots = 0$

Hence $y_2(x) = x^{-1/2} \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right)$, when $a_0 = 1$

$$= \frac{1}{\sqrt{x}} \cos x \text{ is second solution.}$$

$\therefore$ The general solution is

$$y = c_1 y_1 + c_2 y_2$$

$$= \frac{1}{\sqrt{x}} (c_1 \sin x + c_2 \cos x) \text{ (Ans.)}$$

Example-4 Solve Legendre's differential equation $(1-x^2)y'' - 2xy' + n(n+1)y = 0$, when n is a constant.

Solution Given differential equation is

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0 \quad \dots(1)$$

Let the solution of (1) be

$$y = \sum_{r=0}^{\infty} a_r x^{c+r}, a_0 \neq 0 \quad \dots (2)$$

$$\therefore y' = \sum_{r=0}^{\infty} a_r (c+r) x^{c+r-1}$$

$$\text{and } y'' = \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2}$$

Now putting the values of y, y', y'' in (1), we get

$$(1-x^2) \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - 2x \sum_{r=0}^{\infty} a_r (c+r) x^{c+r-1} + n(n+1) \sum_{r=0}^{\infty} a_r x^{c+r} = 0$$

$$\Rightarrow \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - \sum_{r=0}^{\infty} 2a_r (c+r) x^{c+r-1} - \sum_{r=0}^{\infty} 2a_r (c+r) x^{c+r} + (n^2 + n) \sum_{r=0}^{\infty} a_r x^{c+r} = 0$$

$$\Rightarrow \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - \sum_{r=0}^{\infty} a_r [(c+r)^2 - (c+r) + 2(c+r) - n^2 - n] x^{c+r} = 0$$

$$\Rightarrow \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - \sum_{r=0}^{\infty} a_r [(c+r)^2 + (c+r) - n(n+1)] x^{c+r} = 0$$

$$\Rightarrow \sum_{r=0}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - \sum_{r=0}^{\infty} a_r (c+r-n)(c+r+n+1) x^{c+r} = 0$$

$$\Rightarrow a_0 c(c-1) x^{c-2} + a_1 (c+1) c x^{c-1} + \sum_{r=2}^{\infty} a_r (c+r)(c+r-1) x^{c+r-2} - \sum_{r=0}^{\infty} a_r (c+r-n)(c+r+n+1) x^{c+r} = 0$$

$$\Rightarrow a_0 c(c-1) x^{c-2} + a_1 (c+1) c x^{c-1} + \sum_{r=0}^{\infty} a_{r+2} (c+r+2)(c+r+1) x^{c+r} - \sum_{r=0}^{\infty} a_r (c+r-n)(c+r+n+1) x^{c+r} = 0$$

$$\Rightarrow a_0 c(c-1)x^{c-2} + a_1(c+1)cx^{c-1} + \sum_{r=0}^{\infty} \{a_{r+2}(c+r+2)(c+r+1) - a_r(c+r-n)(c+r+n+1)\}x^{c+r} = 0$$

Equating to zero the coefficient of lowest power of x (coefficient of x^{c-2}), then the indicial equation is

$$a_0 c(c-1) = 0$$

$$\Rightarrow c(c-1) = 0 \quad [\because a_0 \neq 0]$$

$$\therefore c = 0, 1$$

Also equating to zero the coefficient of x^{c-1} , we get

$$a_1(c+1)c = 0$$

$$\Rightarrow a_1 = \frac{0}{c(c+1)} \quad [\because a_0 \neq 0]$$

$$\text{For } c = 0, \quad a_1 = \frac{0}{0} \quad [\text{Indeterminate form}]$$

Again equating to zero the coefficient of x^{c+r} , then the recurrence relation is

$$a_{r+2}(c+r+2)(c+r+1) - a_r(c+r-n)(c+r+n+1) = 0$$

$$\Rightarrow a_{r+2} = \frac{(c+r-n)(c+r+n+1)}{(c+r+2)(c+r+1)} a_r \quad \dots (3)$$

When $c = 0$, the recurrence relation (3) becomes

$$\Rightarrow a_{r+2} = \frac{(r-n)(r+n+1)}{(r+1)(r+2)} a_r \quad \dots (4)$$

Now putting $r = 0, 1, 2, 3, 4, 5, \dots$ in (4), we get

$$a_2 = -\frac{n(n+1)}{1 \cdot 2} a_0$$

$$a_3 = -\frac{(n-1)(n+2)}{2 \cdot 3} a_1$$

$$a_4 = \frac{(2-n)(n+3)}{3 \cdot 4} a_2 = \frac{(n-2)n(n+1)(n+3)}{1 \cdot 2 \cdot 3 \cdot 4} a_0$$

$$a_5 = \frac{(3-n)(n+4)}{4 \cdot 5} a_3 = \frac{(n-3)(n-1)(n+2)(n+4)}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} a_1$$

and so on.

From (2), we get

$$y = x^0(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots)$$

$$\Rightarrow y = x^0 \left[a_0 + a_1x - \frac{n(n+1)}{1 \cdot 2} a_0x^2 - \frac{(n-1)(n+2)}{2 \cdot 3} a_1x^3 + \frac{(n-2)n(n+1)(n+3)}{1 \cdot 2 \cdot 3 \cdot 4} a_0x^4 \right. \\ \left. + \frac{(n-3)(n-1)(n+2)(n+4)}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} a_1x^5 + \dots \right]$$

$$\Rightarrow y = a_0 \left[1 - \frac{n(n+1)}{2!} x^2 + \frac{(n-2)n(n+1)(n+3)}{4!} x^4 - \dots \right] \\ + a_1 \left[x - \frac{(n-1)(n+2)}{3!} x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!} x^5 - \dots \right]$$

$\Rightarrow y = Au(x) + Bv(x)$, where A, B are two arbitrary constants and $u(x), v(x)$

represents two solutions.

Example-5 Solve Hermite differential equation $\frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2ny = 0$, where n is a constant.

Solution Try yourself.

Example-6 Solve Laguerre differential equation $x \frac{d^2 y}{dx^2} + (1-x) \frac{dy}{dx} + ny = 0$, where n is a non-negative integer.

Solution Try yourself.

The End